Name: _____

TOPIC TEST 1

Chapter 3: Trigonometry

- Time allowed: 45 minutes
- CAS calculators may be used.
- Part A: 5 multiple-choice questions (5 marks)
- Part B: 8 short answer questions (45 marks)
- Part C: 2 analysis questions (19 marks)

Part A

5 multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 Which statement is NOT true?

A $\sin(180^\circ - x) = \sin(x)$

B $\sin(2\pi - x) = -\sin(x)$

C $\sin(2\pi + x) = \sin(x)$

D $\sin(-x) = \sin(x)$

E $\sin(180^\circ + x) = -\sin(x)$

[1 mark]

2 The compass bearing S 20° E has a true bearing of

A 020°

B 160°

C 110°

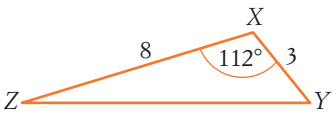
D 200°

E 250°

[1 mark]

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3 Which is the correct statement for the triangle below?



A $YZ = 8^2 + 3^2 - 8 \times 3 \times \cos(112°)$

B $YZ^2 = 8^2 + 3^2 - 8 \times 3 \times \cos(112°)$

C $YZ^2 = 8^2 + 3^2 - 2 \times 8 \times 3 \times \cos(112°)$

D $YZ = 8^2 + 3^2 - 8 \times 3 \times \sin(112°)$

E $YZ^2 = 8^2 + 3^2 - 2 \times 8 \times 3 \times \sin(112°)$

[1 mark]

4 The area of a sector in a circle of radius 5.1 cm, with an angle of 30° subtended at the centre is:

A 6.8 cm^2

B 1.3 cm^2

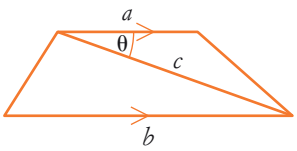
C 390.2 cm^2

D 13.6 cm^2

E 24.8 cm^2

[1 mark]

5 A trapezium has parallel sides a and b. The length of one of its diagonals is c and it makes an angle of θ as shown. Find an expression for its area.



A $A = \frac{1}{2}abc\sin(\theta)$

B $A = ac\sin(\theta)$

C $A = bc\sin(\theta)$

D $A = \frac{1}{2}c\sin(\theta)(a + b)$

E $A = \frac{1}{2}a\sin(\theta)(b + c)$

[1 mark]

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Part B

8 short answer questions
45 marks
Show your working where appropriate.

1 Convert the following to radians, giving exact values.

a 45°

b 300°

c 135°

d 150°

e 400°

[5 marks]

2 Convert the following radians to degrees.

a π

b $\frac{\pi}{3}$

c $\frac{\pi}{6}$

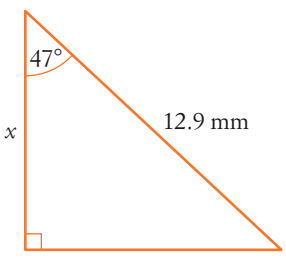
d $\frac{5\pi}{4}$

e 3π

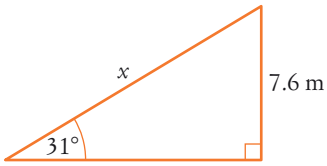
[5 marks]

3 Evaluate x correct to one decimal place.

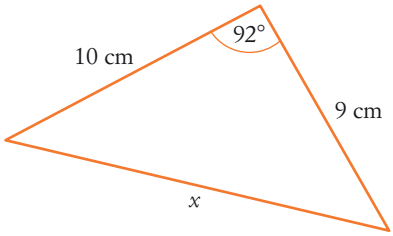
a



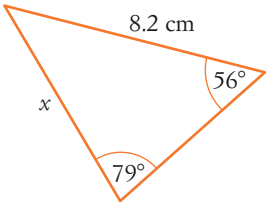
b



c



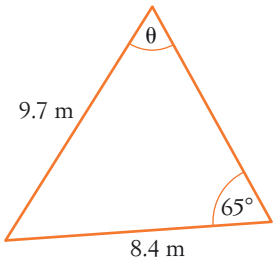
d



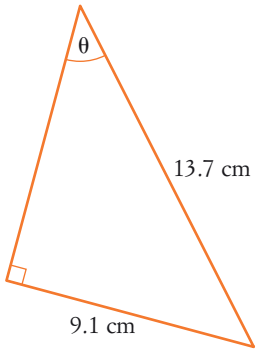
[12 marks]

4 Evaluate θ correct to the nearest degree.

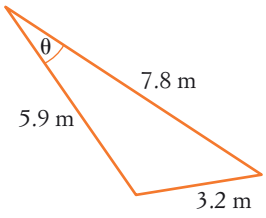
a



b



c



[9 marks]

5 The circumference of a circle is 10 m. Find the length of an arc that is cut off by an angle of $\frac{\pi}{9}$ subtended at the centre of the circle, correct to two decimal places.

[4 marks]

6 A road has a gradient of $\frac{1}{5}$. Find its angle of inclination, correct to one decimal place.

[2 marks]

7 A train leaves Cityview and travels on a bearing of 104° for 137 km. It then changes to another track and travels for 78 km on a bearing of 038° to Harbourtown. What is the straight-line distance between Cityview and Harbourtown?

[5 marks]

8 The angle of elevation up to the top of a tree from a point 12.7 m out from its base is 18°. Find the height of the tree correct to one decimal place.

[3 mark]

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Part C

2 analysis questions
19 marks
Show your working where appropriate.

1 A garden bed is in the shape of a sector of a circle with radius 12 m and the angle at the centre is 30°.

a Find the area of the garden bed, correct to the nearest m^2.

b Fertiliser is needed to fill the garden bed to a depth of 20 cm. What volume of fertiliser is needed?

c A concrete edge is to be built around the outside of the garden bed. It costs $20.50 per linear metre to build this edging. How much will the edging cost in total?

[9 marks]

2 A hot air balloon flying at 950 m/h at a constant altitude of 3000 m is observed to have an angle of elevation of 78°. After 20 minutes, the angle of elevation is 73°. Calculate the angle through which the observer has turned during that time. Answer in degrees and minutes.

[10 marks]
Total marks: 69

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Answers

Part A [1 mark each]

1 $\sin(x)$ is positive in the 1st and 2nd quadrants.

A $\sin(180° - x) = \sin(x)$ is true (2nd quadrant)

B $\sin(2\pi - x) = -\sin(x)$ is true (4th quadrant)

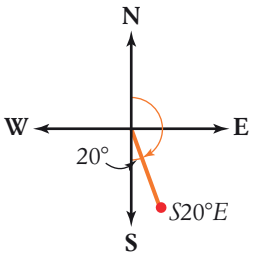
C $\sin(2\pi + x) = \sin(x)$ is true (1st quadrant)

D $\sin(-x) = \sin(x)$ is not true (4th quadrant)

E $\sin(180° + x) = -\sin(x)$ is true (3rd quadrant)

$\therefore$ **D** **[1 mark]**

2 S 20° E is an angle of 20° measured from south and turning towards east.



The true bearing measured clockwise from north is $180° - 20°$ or $160°$.

$\therefore$ **B** **[1 mark]**

3 Cosine rule: $c^2 = a^2 + b^2 - 2ab\cos(C)$

Substituting the information gives $YZ^2 = 8^2 + 3^2 - 2 \times 8 \times 3 \times \cos(112°)$

$\therefore$ **C** **[1 mark]**

4 The formula uses radians.

$30° = \frac{\pi}{6}$ radians

$$A = \frac{1}{2}r^2\theta$$

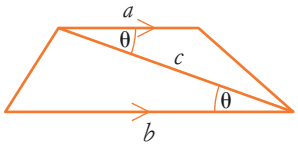
$$= \frac{1}{2} \times 5.1^2 \times \frac{\pi}{6}$$

$$\approx 6.8 \text{ cm}^2$$

$\therefore$ **A** **[1 mark]**

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5 The trapezium is made up of 2 triangles. Since the lines are parallel, θ is in the other triangle (see diagram).



Using $A = \frac{1}{2}ab\sin(C)$ with both triangles:

$$A = \frac{1}{2}ac\sin(\theta) + \frac{1}{2}bc\sin(\theta)$$

Factorising gives $A = \frac{1}{2}c\sin(\theta)(a + b)$

$\therefore$ **D** **[1 mark]**

Part B

1 To convert to radians, multiply by $\frac{\pi}{180^\circ}$ and simplify the fractions.

a $45^\circ \times \frac{\pi}{180^\circ} = \frac{\pi}{4}$ **[1 mark]**

b $300^\circ \times \frac{\pi}{180^\circ} = \frac{5\pi}{3}$ **[1 mark]**

c $135^\circ \times \frac{\pi}{180^\circ} = \frac{3\pi}{4}$ **[1 mark]**

d $150^\circ \times \frac{\pi}{180^\circ} = \frac{5\pi}{6}$ **[1 mark]**

e $400^\circ \times \frac{\pi}{180^\circ} = \frac{20\pi}{9}$ **[1 mark]**

2 To convert to degrees, multiply by $\frac{180^\circ}{\pi}$ or simply replace π by 180°.

a $\pi = 180^\circ$ **[1 mark]**

b $\frac{\pi}{3} \times \frac{180^\circ}{\pi} = 60^\circ$ or $\frac{\pi}{3} = \frac{180^\circ}{3} = 60^\circ$ **[1 mark]**

c $\frac{\pi}{6} \times \frac{180^\circ}{\pi} = 30^\circ$ or $\frac{\pi}{6} = \frac{180^\circ}{6} = 30^\circ$ **[1 mark]**

d $\frac{5\pi}{4} \times \frac{180^\circ}{\pi} = 225^\circ$ or $\frac{5\pi}{4} = \frac{5 \times 180^\circ}{4} = 225^\circ$ **[1 mark]**

e $3\pi \times \frac{180^\circ}{\pi} = 540^\circ$ or $3 \times 180^\circ = 540^\circ$ **[1 mark]**

3 **a** $\cos(47^\circ) = \frac{x}{12.9}$ **[1 mark]**

$12.9\cos(47^\circ) = x$ **[1 mark]**

$8.8 \approx x$

So $x \approx 8.8$ mm. **[1 mark]**

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b $\sin(31°) = \dfrac{7.6}{x}$ **[1 mark]**

$x = \dfrac{7.6}{\sin(31°)}$ **[1 mark]**

≈ 14.8

So $x \approx 14.8$ m **[1 mark]**

c Substitute into the cosine rule.

$c^2 = a^2 + b^2 - 2ab\cos(C)$

$x^2 = 10^2 + 9^2 - 2 \times 10 \times 9 \times \cos(92°)$ **[1 mark]**

$= 187.281\,9094...$ **[1 mark]**

$x = \sqrt{187.2819094}$

≈ 13.7

So $x \approx 13.7$ cm **[1 mark]**

d Substitute into the sine rule.

$\dfrac{a}{\sin(A)} = \dfrac{b}{\sin(B)}$

$\dfrac{x}{\sin(56°)} = \dfrac{8.2}{\sin(79°)}$ **[1 mark]**

$x = \dfrac{8.2\sin(56°)}{\sin(79°)}$ **[1 mark]**

≈ 6.9

So $x \approx 6.9$ cm **[1 mark]**

4 a Substitute into the sine rule.

$\dfrac{\sin(A)}{a} = \dfrac{\sin(B)}{b}$

$\dfrac{\sin(\theta)}{8.4} = \dfrac{\sin(65°)}{9.7}$ **[1 mark]**

$\sin(\theta) = \dfrac{8.4\sin(65°)}{9.7}$ **[1 mark]**

$= 0.784\,843\,856\ldots$ **[1 mark]**

$\theta \approx 52°$ **[1 mark]**

b $\sin(\theta) = \dfrac{9.1}{13.7}$ **[1 mark]**

$\theta \approx 42°$ **[1 mark]**

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c Substitute into the cosine rule.

$$\cos(C) = \frac{a^2 + b^2 - c^2}{2ab}$$

$$\cos(\theta) = \frac{5.9^2 + 7.8^2 - 3.2^2}{2 \times 5.9 \times 7.8}$$ **[1 mark]**

$$= 0.927\,966\,101...$$ **[1 mark]**

$$\theta \approx 22°$$ **[1 mark]**

5 $C = 2\pi r$

$10 = 2\pi r$ **[1 mark]**

$\frac{5}{\pi} = r$ **[1 mark]**

$l = r\theta$

$= \frac{5}{\pi} \times \frac{\pi}{9}$ **[1 mark]**

$= \frac{5}{9}$

≈ 0.56

So the length of the arc is 0.56 m. **[1 mark]**

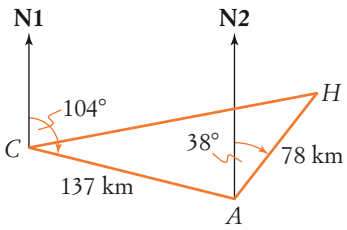
6 Gradient $= \tan(\theta)$

$\frac{1}{5} = \tan(\theta)$ **[1 mark]**

$11.3° \approx \theta$

So the angle of inclination is 11.3°. **[1 mark]**

7 Draw a diagram that shows the information. **[1 mark]**



Find angle CAH.

$\angle CAN_2 = 180° - 104° = 76°$ $(CN_1 \| CN_2)$

$\angle CAH = 76° + 38° = 114°$ **[1 mark]**

Using the cosine rule:

$c^2 = a^2 + b^2 - 2ab\cos(C)$

$CH^2 = 137^2 + 78^2 - 2 \times 137 \times 78 \times \cos(114°)$ **[1 mark]**

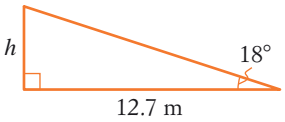
$= 33\,545.775\,54...$ **[1 mark]**

$CH = \sqrt{33\,545.775\,54...}$

≈ 183

So the distance is 183 km. **[1 mark]**

8 Draw a diagram to show the information.



[1 mark]

$$\tan(18°) = \frac{h}{12.7}$$ **[1 mark]**

$$12.7 \tan(18°) = h$$

$$4.1 \approx h$$

The tree is 4.1 m high. **[1 mark]**

Part C

1 a $30° = \frac{\pi}{6}$ radians **[1 mark]**

Area of sector:

$$A = \frac{1}{2}r^2\theta$$

$$= \frac{1}{2} \times 12^2 \times \frac{\pi}{6}$$ **[1 mark]**

$$= 12\pi$$

$$\approx 38 \text{ m}^2$$

So the area of the garden bed is 38 m^2. **[1 mark]**

b 20 cm = 0.2 m

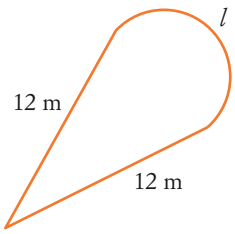
$$V = Ah$$

$$= 12\pi \times 0.2$$ **[1 mark]**

$$\approx 7.5$$

So the volume of fertiliser is 7.5 m^3. **[1 mark]**

c



$$l = r\theta$$

$$= 12 \times \frac{\pi}{6}$$ **[1 mark]**

$$= 2\pi$$

$$\approx 6.3$$

Perimeter of the sector:

$$P = 12 + 12 + 2\pi$$ **[1 mark]**

$$\approx 30.3$$

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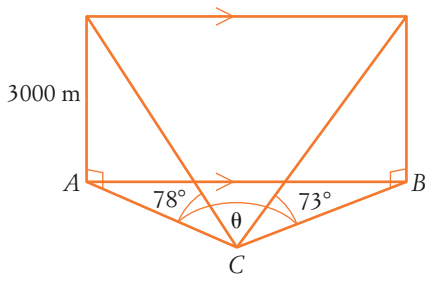
Cost $= 30.3 \times \$20.50$ **[1 mark]**

$\approx \$620.81$

So the cost of the edging is \$620.81. **[1 mark]**

[9 marks]

2 Draw a diagram to show the information.



[1 mark]

Find side AC.

$$\tan(78°) = \frac{3000}{AC}$$

$$AC = \frac{3000}{\tan(78°)}$$ **[1 mark]**

$= 637.669685$ **[1 mark]**

Find side BC.

$$\tan(73°) = \frac{3000}{BC}$$

$$BC = \frac{3000}{\tan(73°)}$$ **[1 mark]**

$\approx 917.192\,044...$ **[1 mark]**

Find side AB.

The balloon travels 950 m in 1 hour, so it travels $\frac{950}{3}$ m or 316.6666… m in 20 minutes.

AB is parallel to the path of the balloon, so $AB = 316.6666…$ **[2 marks]**

Substitute into the cosine rule:

$$\cos(C) = \frac{a^2 + b^2 - c^2}{2ab}$$

$$\cos(\theta) = \frac{637.7^2 + 917.2^2 - 316.7^2}{2 \times 637.7 \times 917.2}$$ **[1 mark]**

$\approx 0.981\,068\,274...$ **[1 mark]**

$\theta \approx 11.2°$ or $11° 10'$ **[1 mark]**

[10 marks]

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